american physiological society
JOURNAL OF APPLIED PHYSIOLOGY

J Appl Physiol 135: 35–52, 2023.
First published May 4, 2023; doi:10.1152/japplphysiol.00070.2023

SYSTEMATIC REVIEW

Physical Activity and the Brain

Does increased core temperature alter cognitive performance during exercise-induced heat strain? A narrative review

Meg A. Doohan,[1,2] Jessica T. Watzek,[3] Neil King,[1,2] Melanie J. White,[1,4] and Ian B. Stewart[1,2]

[1]*Faculty of Health, Queensland University of Technology, Brisbane, Queensland, Australia;* [2]*School of Exercise and Nutrition Sciences, Queensland University of Technology, Brisbane, Queensland, Australia;* [3]*Menzies Health Institute Queensland, Griffith University, Brisbane, Queensland, Australia; and* [4]*School of Psychology and Counselling, Queensland University of Technology, Brisbane, Queensland, Australia*

Abstract

Research to date provides equivocal evidence regarding the influence of heat stress, heat strain, and more specifically, elevated exercise-induced core temperature on cognitive performance. This review sought to identify differences in how specific cognitive tasks were affected by increases in core body temperatures. Included papers (n = 31) measured cognitive performance and core temperature during exercise, while experiencing heightened thermal stress. Cognitive tasks were classified as cognitive inhibition, working memory, or cognitive flexibility tasks. Independently, core temperature changes were not sufficient predictors of cognitive performance. However, reaction time, memory recall, and Stroop tasks appeared to be most effective at identifying cognitive changes during heightened thermal strain. Alterations in performance were more likely to arise under increased thermal loads, which were typically associated with cumulative physiological stressors, such as elevated core temperatures, occurring alongside dehydration, and prolonged exercise durations. Future experimental designs should consider the relevance, or futility of assessing cognitive performance in activities that do not elicit a considerable degree of heat strain, or physiological load.

cognitive performance; core temperature; exercise; heat strain; heat stress

INTRODUCTION

The combined influences of physical activity and heat strain on cognitive performance have long been an area of interest in sporting, military, and occupational settings (1–9). Unfortunately, methodological discrepancies between individual research designs, and the lack of standardization in the use of cognitive tasks, have produced inconsistent and often contradictory findings. Therefore, it has been challenging to ascertain whether the collective effects of heat strain and physical activity does (10–14), or does not (5, 15–17), adversely influence cognitive performance.

Previous evidence has hypothesized that the observable influence of heat strain on cognitive performance is primarily dependent upon the complexity of the task (18). Using this explanation, simpler tasks (reaction time, attentional, inhibition) are more likely to demonstrate improvements, or plateaus in cognitive performance when experiencing elevated heat strain, which is typically identified as occurring at core body temperatures at around 38.5–38.7°C (19). Therefore, the less complex the task is, the greater the likelihood of experiencing benefits from the acute demands placed on the body. Conversely, more complex cognitive tasks (decision making, cognitive flexibility, and executive functioning), which require a greater degree of effort to complete, are anticipated to show notable performance decrements during thermal strain (18–20). This hypothesis proposes that an adverse influence of heat strain on cognitive performance is unlikely to be observed unless the difficulty of the task exceeds a predetermined complexity threshold. However, the practicality of this theory has been questioned in studies incorporating exercise, and/or active heating protocols, given its development using predominantly passive heating procedures (6, 21, 22). Most noticeably, acute bouts of exercise have consistently been documented to facilitate information processing capacities (23), as has moderate exercise-induced hyperthermia (24). Passive models, therefore, cannot solely be used to dictate all cognitive functions during thermal strain. Furthermore, this one-size-fits-all approach seemingly disregards a range of factors that have the potential to influence cognitive performance at varying rates and degrees, such as hydration, body temperature fluctuations, and ambient environmental conditions.

Alternative hypotheses have focused on the notion of cognitive load, where the capacity of brain functions is progressively

Correspondence: M. A. Doohan (meg.doohan@hdr.qut.edu.au).
Submitted 6 February 2023 / Revised 3 May 2023 / Accepted 3 May 2023

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exhausted by the introduction of heat strain (25). With increased thermal load, greater effort is required to maintain optimal cognitive performance (13, 20). Using this hypothesis, only when the degree of heat strain is sufficiently intense, or, when the interaction between stressors is sufficiently high, will decrements in task performance be elicited. Therefore, cognitive performance may initially improve to compensate for an acute increase in stress, and be maintained for an extended period of time—if the overall cognitive load remains relatively stable. However, with increased effort and resources required to preserve these functions, and/or an elevation in the overall load, notable performance impairments are expected to occur (26). Similar to the Maximum Adaptability Model (26, 27), this theory draws on the traditional principles of human stress responses, suggesting that a zone of compensable core temperature values may exist to protect against cognitive performance decrements (26–28). By this model, a slight increase in core temperature may initially provoke improvements in cognitive performance, and preserve performance during moderate, exercise-induced hyperthermia (26–28). Beyond this zone (~38.2–38.7°C), however, where core temperature values progressively exceed compensability, impairments in cognitive task performance are more likely to be observed (27). Although this theory does not discretely acknowledge the influence of additional stressors on cognitive performance, occurring alongside elevated core temperature, the notion of cognitive load does have the potential to highlight such extraneous variables.

The inclusion of several external variables, such as dehydration, physical activity, and extreme environmental conditions, may influence cognitive performance to a degree greater than the independent effects of elevated core temperature. Naturally, however, the interaction of such variables (dehydration, exercise, environmental conditions) makes it difficult to identify the true, and independent effects of elevated core temperature on cognitive performance. Adding to the complexities in determining the true effects of core temperature on cognitive performance, are the methodological inconsistencies prevalent across experimental designs (29–33). Common discrepancies include different environmental conditions, degrees and methods of producing heat strain, physical activity protocols, age, training status, sex, skill levels, and the number, complexity, duration, and repetition of cognitive tasks (34–36). Although such variables have been discussed in earlier reviews (19, 27, 36), more recent assessments have focused on changes in ambient conditions (32, 33) and manipulation of hydration statuses (29, 30) to explain variations in cognitive performance. The results of these reviews have emphasized the potential influence of increased perceptions of effort and decreased thermal comfort in hot environments, which have largely been neglected from the traditional models of physiological strain [i.e., Arousal Theory (inverted-u model), Environmental Determinism Theory] (28, 37).

Using a systematic approach, the purpose of this narrative review was to provide a greater understanding of the tasks and domains of cognitive function that are most impaired with exercise-induced heat strain. Emphasis was placed on determining the presence of task-dependent changes in cognitive function, and whether distinct components of executive function were influenced by different levels of thermal strain (38). Drawing from the hypotheses described, the authors predicted that more complex tasks (cognitive flexibility) would experience the greatest decrements in performance, while simpler tasks (working memory and cognitive inhibition) were not predicted to show any statistically significant changes in cognitive performance. It was also predicted that the degree of heat strain—demonstrated by changes in core temperature experienced during an exercise protocol—would be directly related to the observed variations in cognitive performance.

METHODS

The review was conducted in alignment with the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines (39) and registered on the international prospective register of systematic reviews (PROSPERO Reference: CRD42021226546).

Eligibility Criteria

Studies were considered eligible if they included the following criteria: *1*) human participants, *2*) adults (≥18 yr), *3*) measure of core temperature, *4*) control and experimental groups, *5*) an active method of heat stress (i.e., a physical activity protocol), and *6*) measure of cognitive performance. Studies were required to include both a control and an experimental condition that elicited a change in core temperature at the time of cognitive assessment. Therefore, protocols in which core temperature returned to baseline before conducting cognitive assessments were excluded, and cognitive assessments were required to be completed within the relevant environmental condition. There was no limit on the time or type of exercise conducted. Studies were limited by availability in English, and being a peer-reviewed, empirical experimental design. Studies that did not meet these criteria were excluded from the review.

Studies that included an internal or external method of cooling during an exercise protocol were included in the review, on the basis that all other criteria had been met. Acknowledging that these methods have the potential to significantly alter core temperature, the following three heat manipulation protocols were classified. Standard protocols were identified as designs where cold, or thermoneutral environments acted as the control, and hot environments as the experimental condition. Internal cooling mechanisms incorporated a form of menthol, ice, or liquid ingestion to alter core temperature, either before or during exposure to heat strain. In these studies, the cooling conditions were used as the control, due to the reduced core temperature values, and the noncooling trials as the experimental conditions. Finally, external mechanisms of inducing heat strain included wearing some form of personal protective equipment (PPE), where the control condition was the trial in which the PPE was not donned. In this same category, tasks that used external cooling vests were also permitted, wherein the conditions were reversed; control conditions occurred when participants were wearing the cooling apparatus.

Search Strategy

Three online databases were used for the search: Cumulative Index of Nursing and Allied Health Literature (CINAHL), Embase, and PubMed. The search consisted of three major

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themes: *1*) physical activity, *2*) core temperature, and *3*) cognitive performance. Terms were adjusted for each database by applying mesh terms and filters (see APPENDIX). Pilot searches occurred in October 2021 to ensure robustness of the search strategy, and the final search was conducted on May 11, 2022.

Data Extraction

All studies were imported into Endnote (v.X9, 2013), before being uploaded onto Covidence (v2715, 2021) to complete the extraction. After removing duplicates, studies, titles, and abstracts, followed by full texts, were screened independently by two reviewers (M.D., J.W.) using standardized criteria. Disagreements were solved through discussions between reviewers, and if necessary, a third reviewer (I.S.) was introduced to resolve conflicts. Where possible, extracted data included: participant demographics [age, maximal oxygen consumption ($\dot{V}O_{2max}$), sex, sample size, training status, hydration manipulation, acclimatization status, fuel intake, external weight bearing], experimental design, exercise protocol (type, intensity, and time), environmental manipulation, mean (*M*) and standard deviation values (SD) of cognitive assessments, and core temperature values from matched time points. To assess changes in core temperature across studies, the time point and experimental condition at which the peak core temperature change occurred, when compared with the control condition, were extracted. Any required data points that were only displayed graphically were digitized using WebPlotDigitiser (v.4.4).

Quality Assessment

Quality assessment analyses were conducted by two reviewers (M.D., I.S.) using the Cochrane Risk of Bias Tool (40) and appropriate templates for parallel and crossover trials. The assessment aided in quantifying potential risks in allocating participants to conditions, standardizing experimental protocols, missing data, condition blinding, and statistical analyses.

Statistical Analysis

Qualitative analyses aimed to identify similarities between research designs and results of specific tasks or domains, and to provide a foundation of which to inform future design methodologies. Tasks were categorized under three discrete domains, derived from Diamond (41): cognitive inhibition, working memory, and cognitive flexibility. Studies were further identified as being either a between-groups or repeated-measures design. Modes of heat strain were classified as external (cooling vests, heat pads, clothing), internal (fluid, ice consumption), or standard (thermoneutral/cold vs. hot environments). Each task was categorized into the relevant domain, based on previous literature, or author classifications (41). Cognitive outcomes such as reaction time (milliseconds; ms), accuracy (percentage; %), and errors (number, %), as well as the unit of measure (ms, % of in/correct responses), were extracted for analyses. Hydration and nutrition, acclimatization status, load carriage, training status, and clothing were also considered as potentially influential variables. Using the time point and condition at which the peak difference in core temperature occurred between control and experimental groups, and the associated cognitive outcome, effect size was calculated using Hedge's *g* (42)—identifying the influence of heat strain on performance of a cognitive task, using $P \leq 0.05$ significance. A meaningful change in cognitive performance was determined where the 95% confidence intervals around the Hedge's *g* value did not cross zero. Where a meaningful change was observed, this implied that there was a statistical difference between conditions. Although it is difficult to conclude whether this would transfer to a meaningful effect on cognitive performance in the field; within competitive athletic, or life-death contexts, any detrimental impact could be considered meaningful, and so statistical difference provided some assurance that this would be a reliable impact in such situations.

RESULTS

Search Results

The final search produced 866 results; 751 when duplicates were removed. Of these, 671 studies were removed after title and abstract screening. Full-text reviews were conducted on the remaining 80 studies, plus an additional five studies derived from gray literature (n = 85). Fifty-four studies were removed due to not having control conditions, not measuring cognitive performance when core temperature was different between conditions, or not incorporating an exercise protocol. From the remaining 31 studies, there were six incomplete datasets (24, 43–48). However, changes in core temperature were able to be extracted from two of these studies (45, 47) and so 27 studies were incorporated in the quantitative analyses of core temperature. The PRISMA diagram in Fig. 1 describes the process of record selection.

Quality Analysis Assessment

Using the Cochrane Risk of Bias Tool (40), reviewers identified that despite some differences between categories, all studies returned a moderate risk of bias result. The adherence to protocol intervention and reporting of results were the two domains that were most subject to bias. Given the typical inability to blind participants to experimental conditions in thermoregulatory research, and as many of the studies included multiple time points and methods of calculation and analysis for cognitive tasks, these results are not surprising. Nevertheless, it is imperative to acknowledge such risks when analyzing and interpreting the results of each study. Table 1 displays the results of the quality assessment analysis.

Study Characteristics

There was a total of 499 participants within the 31 studies (M = ~16), including 41 females in eight studies. Using available data, participant demographics were: Age: M = 24.03, SD = 3.18 yr, $\dot{V}O_{2max}$: M = 52.8, SD = 5.6 $mL^{-1}\cdot kg^{-1}\cdot min$. Repeated-measures designs were used in 25 studies, whereas five studies used a between-groups design (2, 43, 49–51), and in one, an intervention design (63). Training status was unknown in two studies (46, 51) with participants either recreationally (n = 18) or well-trained (n = 11) in the remaining studies.

Acclimatization status was unidentified in 15 studies. In 14 studies, participants were unacclimatized, and two included acclimatized participants (2, 63). External loading was not manipulated in 26 studies. In the remainder, participants

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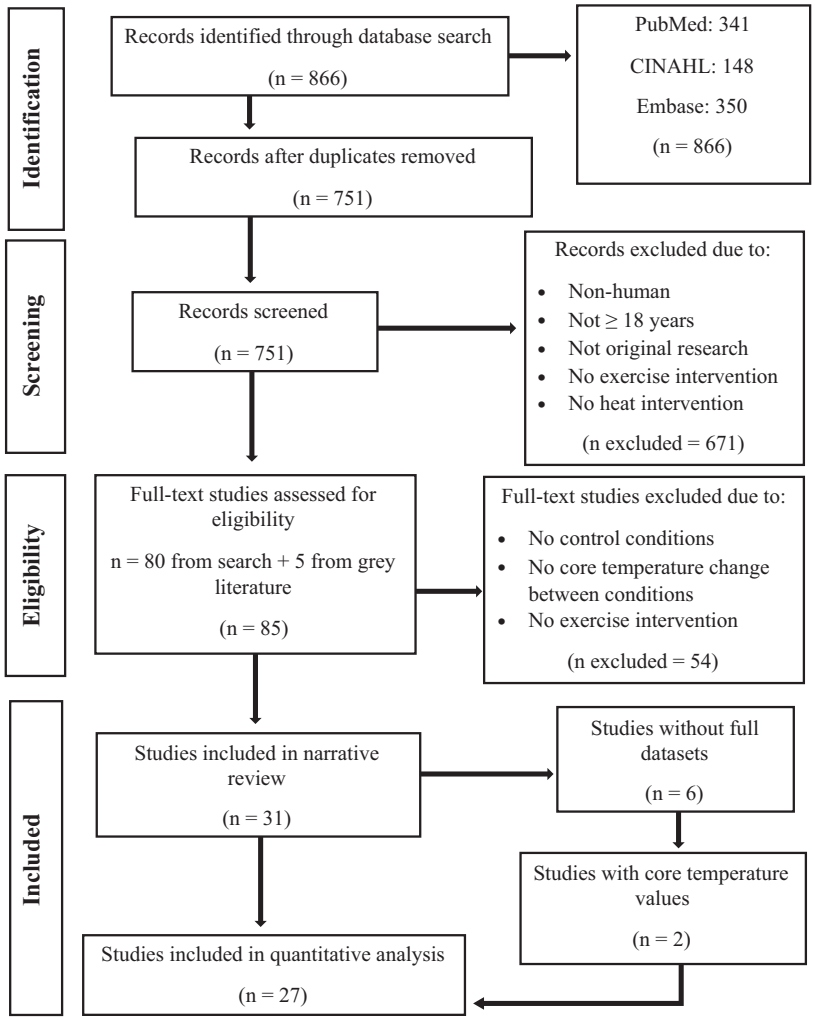

Figure 1. Preferred reporting items for systematic review and meta-analysis (PRISMA) diagram representing the process of study screening for eligibility of qualitative review and effect size analysis. *n* refers to number of records/articles.

wore 20-kg backpacks (2, 52), lifted 22.7-kg sandbags (45), carried 20% of body mass (51), or walked with a 22.7-kg backpack (49). Similarly, 26 studies did not manipulate food intake; one study incorporated scopolamine consumption (43), two included sports drinks (electrolytes) (24, 49), and two examined carbohydrate consumption (55, 58). Clothing was not identified in 12 studies; eight studies used personal protective gear, in ten, athletic clothing, and in one, long pants and shirts were worn (49). Hydration status was not manipulated in 18 studies. One study allowed ad lib consumption of fluid throughout the experimental procedures (49), whereas the remaining 12 studies used various forms of water provision and/or restriction to alter hydration statuses between control and experimental groups (6, 15, 21, 24, 46, 48, 52, 54, 56, 62, 64, 65).

Walking protocols were used in 14 studies, whereas cycling and running were used in six each, and team sports in five. Eleven studies used standard environments to manipulate core temperature, five used internal methods, and two used external mechanisms (3, 56). Five studies used a combination of internal and standard mechanisms, four included external and standard methods (17, 44, 53, 57), and four used internal and external protocols (21, 48, 58, 62). Table 2 presents the descriptive characteristics of all studies.

Cognitive Performance

Cognitive tasks were categorized into one of the three aforementioned domains: cognitive inhibition, working memory, and cognitive flexibility (41). Cognitive inhibition was the most popular domain, being used in 24 of the 31 studies. Tasks of working memory were used in 11 different studies, whereas cognitive flexibility was assessed within 14 separate experimental procedures. Cognitive data was not available for comparison in six of the 31 studies (24, 43–47). There were also three incomplete datasets (56, 57, 64); only the available data in these studies were analyzed. Given the large range of cognitive tasks used, results between, and within studies, and tasks, were mixed.

Core Temperature Changes

Effect sizes (Hedges's *g*) were generated for each cognitive task at the time point in the study that elicited the peak difference in core temperature between the control and experimental conditions (Tables 3 and 4), using the *dplyr* package in R Studio (v.1.3.1056). Despite attempts to contact authors, restrictions on data sharing meant that four of the 31 studies could not have their peak changes in core temperature identified (24, 43, 44, 46). From the 27 studies where core temperature values were extracted, 18 evoked small core temperature

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Table 1. *Cochrane risk of bias assessment for n = 31 studies*

Between-Groups Studies						
Reference	Domain 1: Randomization Process	Domain 2: Deviations from Intended Interventions	Domain 3: Missing Outcome Data	Domain 4: Measurement of Outcome	Domain 5: Selection of Reported Result	Overall Risk of Bias
Deming et al. (49)	?	?	?	×	?	?
Maric et al. (50)	✓	?	✓	✓	?	?
Parker et al. (51)	?	?	✓	✓	?	?
Radakovic et al. (2)	?	?	✓	✓	?	?
Safer (43)	?	×	?	?	?	?

Within-Groups Studies							
Reference	Domain 1: Randomization process	Domain S: Period and Carryover Effects	Domain 2: Deviations from Intended Interventions	Domain 3: Missing Outcome Data	Domain 4: Measurement of Outcome	Domain 5: Selection of Reported Result	Overall Risk of Bias
Adams et al. (52)	×	✓	×	✓	?	×	?
Aljaroudi et al. (53)	✓	?	?	✓	?	×	?
Ashworth et al. (54)	✓	✓	?	✓	?	?	?
Bailey et al. (55)	✓	?	?	✓	✓	?	?
Bandelow et al. (24)	×	✓	?	×	?	×	?
Benjamin et al. (48)	?	?	×	×	×	×	?
Benor and Shvartz (44)	?	?	×	×	?	×	?
Caldwell et al. (56)	×	?	×	×	?	×	?
Caldwell et al. (57)	×	✓	×	?	?	×	?
Clarke et al. (58)	?	?	?	✓	?	×	?
Clarke et al. (17)	?	?	×	✓	?	×	?
Coehoorn et al. (3)	?	?	×	✓	?	×	?
Donnan et al. (59)	✓	✓	?	✓	?	?	?
Edwards et al. (60)	?	✓	×	?	?	?	?
Ganio et al. (46)	✓	✓	?	×	✓	×	?
Gerhart et al. (45)	×	✓	?	?	?	×	?
Macleod et al. (6)	✓	✓	×	×	?	×	?
Malan et al. (61)	?	?	?	✓	?	?	?
Mazalan et al. (62)	✓	×	?	×	?	?	?
Saldaris et al. (21)	?	✓	?	?	?	?	?
Saldaris et al. (15)	?	?	?	✓	?	?	?
Shibasaki et al. (22)	?	✓	?	✓	?	×	?
Tamm et al. (63)	×	✓	×	✓	×	×	?
Tikuisis and Keefe (64)	×	✓	?	✓	?	×	?
Watkins et al. (47)	?	✓	×	×	?	?	?
Wittbrodt et al. (64)	?	✓	?	✓	?	?	?

Note: Tick, low risk of bias; question mark, some concerns of bias; cross, high risk of bias.

changes (between 0°C and 1.0°C) (3, 6, 15, 17, 21, 22, 45, 47, 48, 51, 53, 56, 58–62, 64), seven induced moderate changes (between 1.0 and 2.0°C (49, 52, 54, 55, 57, 63, 65), and two produced large body temperature changes, greater than 2.0°C (2, 50).

Tables 3 and 4 provide preliminary evidence of the influence individual experimental protocols had on cognitive performance—as reflected by the effect sizes, and the respective confidence intervals around the effect sizes not encompassing zero. Categorization by core temperature change orders studies from those that showed the smallest core temperature changes, to those with moderate to large core temperature changes between conditions. Table 3 portrays response outcomes as measured by accuracy, whereas Table 4 depicts reaction time outcomes. Both tables also categorize specific tasks into the separate cognitive domains (cognitive inhibition, working memory, and cognitive flexibility).

DISCUSSION

The initial hypothesis predicted that more complex tasks (cognitive flexibility) would show the greatest decrements in performance, compared with simpler tasks (working memory, cognitive inhibition), with elevations in core temperature. This review, however, was unable to support the hypothesis. Confounding issues included a large number of inconsistencies between research designs, methodologies, and task outcome measures. Preliminary evidence did indicate that moderate levels of core temperature change (>1°C) were at least required to influence cognitive performance across reaction time, motor screening, information processing, recognition, altered go/no-go, standard memory, list recall, vigilance, classic and numerical Stroop, and three-term reasoning tasks. Furthermore, although the greatest performance decrements typically occurred when core temperatures reached ≥39°C, of the outcome measures evaluated, irrespective of cognitive domain, task accuracy was more detrimentally affected compared with task response time.

In support of a recent review (27) of the Maximum Adaptability Model, in studies in which the maximum core temperatures of the examined conditions did not exceed 39°C (23 of the 31 articles reviewed) and/or the difference in core temperature between studies was less than 1°C (18 articles), predominantly no differences in cognitive

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performance were observed. Impairments were more likely to be observed when heat strain was combined with additional physiological stressors, such as prolonged bouts of exercise, extended periods of time spent at elevated core temperatures, and increased levels of dehydration. For example, of the two studies that elicited greater than 2°C change in core temperature combined with 90-min of exercise, both produced observably negative changes in performance (2, 50).

Cognitive Inhibition

Tasks of cognitive inhibition require a series of attentional control and inhibitory processes to override impulsive behaviors and produce the most desirable responses. Among the simplest of the three domains examined, such assessments typically include response and reaction time, vigilance, and target detection tasks (41). Within the current review, 17 distinct tasks of cognitive inhibition were included across 20 experimental designs. Of these tasks, no changes in either performance outcome was identified within any choice reaction time (15, 21, 48), numerical vigilance (47), dual task tracking (47), visual vigilance (52, 59), time perception (65), shape recognition (56, 57), response time (61), perceptual processing (54), virtual environment (55), and standard go/no-go (3, 17, 22, 53, 54) tasks. A debilitating effect of heat was observed in a rapid visual information processing task (2), whereas a positive influence of heat was noted within a single shape and shade recognition task (57). Contradictory evidence was documented in standard vigilance (54), modified go/no-go (65), psychomotor vigilance (51), motor screening (2, 50), and reaction time (2, 50, 56, 57) tasks.

Accuracy responses were more detrimentally affected than response time outcomes in tasks of cognitive inhibition. In interventions that elicited >1°C, only 12% of tasks showed decrements in response time, compared with 28% for accuracy responses, whereas designs that evoked <1°C change resulted in only 5% of tasks showing significant effects of heat in cognitive outcomes. However, caution should be observed in dismissing tasks that have only been investigated once or where minimal physiological stress has been induced.

A detrimental effect of heat on both accuracy and response time outcomes was observed after 90-min of walking that provided a large core temperature change (>2°C), in a rapid visual information processing task, alongside an appropriately powered design (n = 40) (2). Within the same study, a negative influence of heat was also observed in motor screening and reaction time tasks for accuracy, but not response time outcomes. Using an identical experimental design (50), or an extended walking bout (56), similar results were observed for a reaction time task. These large changes in core temperature and/or extended exercise durations produce significant physiological load, and so are potentially more likely to demonstrate noticeable changes in performance in tasks of cognitive inhibition. On the contrary, in a single study (54), a negative effect on a vigilance task was observed in response time, but not in accuracy responses. However, this was in only one of seven outcomes within the study. Using a passive preheating protocol to elevate core temperature before the exercise bout, the results of this study highlight the importance of identifying the true, and combined effects that heat and exercise have on cognitive performance. Nevertheless, not all outcomes demonstrated negative effects of heat on cognitive inhibition. The following studies (51, 57, 65), while documenting some positive changes of heat and exercise on cognitive inhibition performance, did not induce core temperature elevations beyond 38.5°C.

In summary, the standard reaction time task produced the most consistent performance decrements in terms of accuracy, across several experimental designs. While the go/no-go task appears to be the least influenced, this was predominantly utilized in studies that induced only small-to-moderate changes in core temperature. Consistent negative results of heat on accuracy performance were produced across multiple cognitive inhibition tasks when large changes in core temperature were combined with extended exercise bouts. Such results highlight how imperative it is to acknowledge the influence of cumulative stressors on cognitive performance, and overall stress loads experienced by participants, when designing, and analyzing experiments.

Working Memory

Working memory performance is reliant on an individual's ability to hold information, and manipulate it, as is prominent in tasks requiring delayed recall and recognition, visual searches, and object and numeral spans (41). Typically, working memory tasks require an increased degree of complexity from simple inhibitory tasks. In the present analysis, 12 discrete tasks were used across eight studies. Of these tasks, only two showed any changes in accuracy, but not response time performance—a memory (54) and word list recall (49) task. No significant changes were observed for visual search (6), mathematical pairs (58), digit span (54), pattern recognition (50), match-to-sample visual search (50), serial sevens (15, 62), Sternberg (6), letter-digit recognition (64), complex span (OSPAN) (62), or pattern comparison (64) tasks.

There were no significant changes in response time performance for any working memory tasks (6, 50, 64). The Sternberg (6), match to sample visual search (50), and pattern recognition (visual and spatial) memory (50) tasks, also documented no significant changes in accuracy responses at the examined time point. However, only one of these studies documented a large body temperature change, and extended exercise duration, resulting in absolute core temperature values exceeding 39°C (50). While noting that these results only occur within a single study, the cumulative physiological load experienced in this design potentially indicates that the tasks used in this study are less susceptible to exercise-induced heat strain.

Accuracy remained unchanged in two serial sevens tasks (15, 62), and single visual search (at both baseline and complex levels) (6), mathematical pairs (58), OSPAN (62), and digit span (54) tasks, also indicating that these tasks may be less susceptible to exercise-induced heat strain. Specifically, while the digit span task did not show any changes in accuracy performance, a memory task (involving recall of information about a previously presented map) used in the same study was shown to be negatively affected by heat. However, singular uses of the remaining three tasks prevents definitive conclusions being reached, as the experimental designs only induced small changes in core temperature (6, 15, 58, 62).

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Table 2. *Descriptive results of all (n = 31) studies extracted*

Reference	*n* (M/F)	Age *M* (SD)	$\dot{V}O_{2max}$ *M* (SD)	Groups	T_{core} Measure	Peak T_{core}, °C	Peak T_{core}, °C, Change (Time)	Exercise Protocol	Environment °C (% RH)
Adams et al. (52)	12 (12/0)	*20* (2)	*57* (6)	**Euhydrated temperate (C)**	Rectal	37.86	1.46 (90 min)	90 min walk @ 50% $\dot{V}O_{2max}$	**TEMP**: *18.0* (50)
				Euhydrated hot		38.25			**HOT**: *34.0* (45)
				Hypohydrated temperate		38.22			
				Hypohydrated hot (E)		39.32			
Aljaroudi et al. (53)	12 (12/0)	*24* (3.2)	*56.3* (7.4)	**Cooling (C)**	Rectal	37.80	0.50 (40 min)	40 min running @ 40% $\dot{V}O_{2max}$	*30* (70)
				No cooling (E)		38.30			
Ashworth et al. (54)	8 (3/5)	*28.13* (4.82)	—	**Control (C)**	Rectal	37.61	1.08 (15 min)	30-min steady-state walking	**CONT**: 36 (75), 33°C WI
				Menthol		38.64			**HOT**: *40* (75), 33°C WI
				Hot (E)		38.68			
Bailey et al. (55)	10 (10/0)	*23* (1)	*49.3* (1.4)	**Temperate placebo (C)**	GI	37.91	1.04 (>101 min)	Cycling @ 80% $\dot{V}O_{2max}$ until fatigue (between 101 and 169 min)	**TEMP**: *22* (50)
				Temperate CHO		37.95			**HOT**: *35* (70)
				Hot placebo		38.52			
				Hot CHO (E)		38.59			
Bandelow et al. (24)	20 (20/0)	*20.2* (2)	—	Match 1—ad lib	GI			90 min football game	**1**: *34.3* (64)
				Match 2—hydration					**2**: *34.4* (65)
				Match 3—cooling					**3**: *33.8* (62)
Benjamin et al. (48)	12 (12/0)	*20* (1)	*53.9* (7.3)	Euhydrated (No dousing)	Rectal	38.47	0.92 (75 min)	5 × 15 min treadmill run	*34.7* (46)
				Euhydrated (Dousing) (C)		38.36			
				Hypohydrated (No dousing) (E)		39.28			
				Hypohydrated (Dousing)		39.01			
Benor and Shvartz (44)	7 (7/0)	—	—	Temperature only (×5)	GI	—	—	120 min continuous walking	**TEMP + COOLING**: *30, 35, 40, 45, 50*
				Cooling garment (×5)					
Caldwell et al. (56)	9 (9/0)	*27.3* (5.43)	—	**Control (C)**	Tympanic	37.85	0.43 (150 min)	150-min steady-state walking at varied intensity	*36* (60)
				Torso armour		38.00			
				Full armour (E)		38.27			
Caldwell et al. (57)	8 (8/0)	*27.1* (6.2)	—	**Control (C)**	Tympanic and rectal	36.92	1.43 (75 min)	8 × 13 min cycling @ ~30 W, 2 min rest bouts	**CONT**: *20* (30), **HOT**: *48* (20)
				Experimental (E)		38.36			
				Auxiliary cooling		37.28			

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Table 2.— *Continued*

Reference	*n* (M/F)	Age *M* (SD)	$\dot{V}O_{2max}$ *M* (SD)	Groups	T_{core} Measure	Peak T_{core}, °C	Peak T_{core}, °C, Change (Time)	Exercise Protocol	Environment °C (% RH)
Clarke et al. (58)	12 (12/0)	*25* (1)	*61.3* (1.4)	**Pre-cooling placebo (C)**	GI	38.96	0.53 (15 min)	2 × 45 min walk, jog, sprint intervals	*30.5* (0.2)
				Pre-cooling CHO		38.63			
				Heat placebo (E)		39.06			
				Heat CHO		39.05			
Clarke et al. (17)	8 (8/0)	*28* (6)	*53* (6)	**Pre-cooling (C)**	Rectal	37.14	0.71 (30 min)	2 × 45 min steady-state running	*32.4* (6.4)
				No cooling (E)		37.86			
Coehoorn et al. (3)	15 (15/0)	*32.7* (12.2)	*51.3* (6.7)	**No gear (C)**	GI	38.45	0.65 (>50 min)	120 min graded intensity walking	*25–56*
				Gear (E)		39.10			
Deming et al. (49)	24 (18/6)	*29* (3)	*51.8* (1.8)	**Normothermic water (C)**	GI	37.66	1.13 (120 min)	120 min walking @ 4.8 km/h, 7% grade	*21*
				Heat water		38.73			*33* (10)
				Heat electrolytes		38.56			
				Heat electrolytes + CHO (E)		38.79			
Donnan et al. (59)	12 (12/0)	*21.4* (3.3)	*53*	**Control (C)**	GI	37.91	0.48 (40 min)	2 × 20 × 2 min cycle: 5 sec max, 105 s ~35% $\dot{V}O_{2max}$, 10 sec rest	**CONT**: *18.0* (51.9), **HEAT**: *31.6* (49.3)
				Heat (E)		38.39			
Edwards et al. (60)	11 (11/0)	*24.4* (3)	*50.91* (2)	**Fluid intake (C)**	GI	38.67	0.43 (95 min)	45 min cycle, 45 min team game	*24–25* (47–55)
				No fluid (E), Mouth rinse		39.01			
						38.91			
Gerhart et al. (45)	10 (8/2)	*24* (2.5)	*49.7* (15.8)	**Thermoneutral (C)**	GI	37.92	0.10 (30 min)	2 × 10 min walk @ 70-75% MHR, w/2× 15 22.7 kg lifts	**THER**: *25* (40), **HEAT**: *37.8* (60)
				Heat (E)		38.02			
Ganio et al. (46)	26 (26/0)	*20* (0.3)	—	Exercise + diuretic	GI	—	—	3 × 40 min walking bouts	*27.7* (42)
				Exercise + placebo					
				Euhydration + placebo					
Macleod et al. (6)	8 (0/8)	*22* (3)	*53.4* (2.2)	**Thermoneutral fluid (C)**	Rectal	38.01	0.52 (50 min)	2 × 25 min running bouts	**THER**: *16* (53)
				Thermoneutral no fluid		38.32			**HOT**: *33* (59)
				Hot fluid (E), hot no fluid		38.53			
						38.51			

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Table 2.— *Continued*

Reference	*n* (M/F)	Age *M* (SD)	$\dot{V}O_{2max}$ *M* (SD)	Groups	T_{core} Measure	Peak T_{core}, °C	Peak T_{core}, °C, Change (Time)	Exercise Protocol	Environment °C (% RH)
Malan et al. (61)	4 (2/2)	*22* (3)	—	**Cool (C)**	GI	37.51	0.25 (70 min)	Mimic goal keeper. 2 × 35 min, 10 min rest	**COOL**: *20* (40)
				Hot (E)		37.76			**HOT**: *35* (40)
Maric et al. (50)	40 (40/0)	*20* (0.9)	*53.4* (8.28)	**Hot (E)**	Tympanic	39.50	2.72 (90 min)	90 min walking @ 5.5 km/h	**HOT**: *40*, 39°C WBGT
				Control (C)		36.78			**CON**: *20*, 16°C WBGT
Mazalan et al. (62)	10 (10/0)	*26.1* (1.9)	*53.9** (4.3)	**Head cooling (HC) (E)**	GI	39.57	0.46 (60 min)	2 × 30 min running bouts @ 70% of $\dot{V}O_{2peak}$	*35* (70)
				Head cooling + ice Ingestion (C)		39.13			
				Room temp water ingestion		39.54			
Parker et al. (51)	40 (24/16)	*27.75*	—	**Temperate (C)**	GI	37.7	0.50 (90 min)	90 min walk @ 40–45% $\dot{V}O_{2max}$	**TEMP**: *22–24*
				Hot (E)		38.2			**HOT**: *35–38*
Radakovic et al. (2)	40 (40/0)	*20.1* (0.9)	*58.17* (7.2)	**Unacclimated cold (C)**	Tympanic	37.80	2.61 (90 min)	90 min walking @ 5.5 km/h	**COLD**: *20*, 16°C WBGT
				Unacclimated hot (E)		39.40			**HOT**: *40*, 29°C WBGT
				Passively acclimated hot		39.20			
				Actively acclimated hot		38.59			
Safer (43)	48 (48/0)	—	—	Cold drug + work	—	—	—	10 min walk per half hour for 6 h	**COLD**: *4.44* (40)
				Cold drug					**THER**: *21.11* (40)
				Cold work					**MOD**: *29.44* (40)
				Thermoneutral drug + work					**HOT**: *35* (40)
				Thermoneutral drug					**SEVERE**: *40.56* (40)
				Thermoneutral work					
				Moderate drug + work					
				Moderate drug					
				Moderate work					
				Hot drug + work					
				Hot drug					
				Hot work					
				Severe drug					
				Severe temp only					
Saldaris et al. (21)	10 (10/0)	*23.1* (2.4)	*48.5* (3.6)	**ICE (C)**	GI	37.73	0.33 (15 min)	60 min cycling @ 55% of $\dot{V}O_{2peak}$	*35* (50.2)
				Water (E)		37.93			
Saldaris et al. (15)	12 (12/0)	*25.3* (4.2)	*61.3** (4.3)	**Crushed ice + menth (C)**	GI	38.17	0.20 (30 min)	3 × 30 min run @ 65% $\dot{V}O_{2peak}$	*35.3* (59.2)

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Table 2.— *Continued*

Reference	*n* (M/F)	Age *M* (SD)	$\dot{V}O_{2max}$ *M* (SD)	Groups	T_{core} Measure	Peak T_{core}, °C	Peak T_{core}, °C, Change (Time)	Exercise Protocol	Environment °C (% RH)
				Water + menthol		38.38		TTF run @ 100%	
				Water (E)		38.37			
Shibasaki et al. (22)	15 (15/0)	*20.8* (0.9)	—	**Temperate (C)**	Tympanic	37.19	0.67 (60 min)	4 × 15 min cycling bouts @ 67% $\dot{V}O_{2max}$	**TEMP**: *20* (30–40)
				Hot (E)		37.86			**HOT**: *35* (30–40)
Tamm et al. (63)	20 (20/0)	*24.9* (3.7)	*53.8* (7.1)	**Thermoneutral (C)heat, pre-HA (E)**	Rectal	38.20	1.50 (90 min)	Walking @ 60% $\dot{V}O_{2max}$ until fatigue (between 85 and 158 min)	**THER**: *22* (35)
				Heat, post HA		39.70			**HEAT**: *42* (18)
						39.70			
Tikuisis and Keefe (65)	11 (9/2)	*28.9* (6)	—	**Control (C)**	Rectal	37.35	1.12 (180 min)	30 min walking bouts for max 240 min	**CONT**: *22*
				Heat hydration		38.41			**HOT**: *28–30* (42°C water perfused)
				Heat dehydration (E)		38.47			
Watkins et al. (47)	13 (13/0)	*19.6* (3)	—	**Cold (C)**	Rectal	37.15	0.36 (90 min)	2 × 45 min football simulation	**COLD**: *−5* (50)
				Thermoneutral		37.16			**THER**: *18* (50)
				Heat (E)		37.19			**HOT**: *30* (50)
Wittbrodt et al. (64)	12 (12/0)	*22.2* (2.4)	*42.8* (4.8)	**Fluid replacement (C)**	Rectal	37.76	0.46 (50 min)	50 min cycling @ 60% of $\dot{V}O_{2peak}$	32°C, 65% RH
				No fluid (E)		38.22			
				Ad libitum		37.86			

Conditions and studies in bold are those used within the quantitative analyses of effect sizes and experimental groups used for these analyses. Studies in italics indicate there is missing cognitive data. (C), control condition; (E), experimental condition; GI, gastrointestinal temperature; M/F, male/female; Max, maximal effort; MHR, maximal heart rate; Min, minutes; RH, relative humidity; Sec, seconds; T_{core}, core temperature; TTF, time to fatigue; $\dot{V}O_{2max}$, maximal oxygen consumption; $\dot{V}O_{2peak}$, peak oxygen consumption; WI, water immersion; —, unavailable data. Peak T_{core} recorded at time of cognitive measurement. *Peak $\dot{V}O_2$ measure.

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Table 3. *Cognitive task results and core temperature changes for accuracy response (displayed in ascending core temperature difference, for each cognitive domain)*

Reference	Peak T_{core} Difference, °C	Time Point	Cognitive Task	Performance Measure	Unit of Measure	Influence of Heat	Hedges' *g*	Lower CI	Upper CI
				Cognitive Inhibition					
Saldaris et al. (15)	0.20	30	Choice reaction time	Accuracy	Percentage correct		0.25	−0.55	1.05
Watkins et al. (47)	0.36	90	Numerical vigilance	Targets hit	Number correct		—	—	—
				Missed targets	*Number incorrect		—	—	—
				False targets	*Number incorrect		—	—	—
			Dual task tracking	Accuracy	Percentage correct		—	—	—
				Missed targets	*Number incorrect		—	—	—
				False targets	*Number incorrect		—	—	—
Caldwell et al. (56)	0.43	150	**Reaction time**	**Accuracy**	**Number correct**	**Negative**	**−1.70**	**−2.82**	**−0.58**
Donnan et al. (59)	0.48	40	Visual vigilance	Accuracy	Percentage correct		−0.31	−1.12	0.49
Aljaroudi et al. (53)	0.50	40	Go/no-go	Accuracy	*Number incorrect		0.13	−0.67	0.93
Parker et al. (51)	0.50	90	Psychomotor vigilance test	Minor Attention Lapse	*Number incorrect		−0.3	−0.93	0.32
				False Starts	*Number incorrect		−0.27	−0.89	0.35
Coehoorn et al. (3)	0.65	>50	Go/no-go	Accuracy	*Number incorrect		3.83	2.57	5.09
Shibasaki et al. (22)	0.67	60	Go/no-go	Accuracy	Percentage correct		0.01	−0.71	0.72
Clarke et al. (17)	0.71	30	Go/no-go	Accuracy	Percentage correct		0.2	−0.78	1.19
Bailey et al. (55)	1.04	>101	Virtual environment	Kills	Number correct		0.59	−0.31	1.49
				Failures	*Number incorrect		0	−0.88	0.88
Ashworth et al. (54)	1.08	15	Go/no-go	Accuracy	Percentage correct		−0.19	−1.17	0.8
			Perceptual processing	Accuracy	Percentage correct		−0.84	−1.86	0.21
			Vigilance	Maximum span	Levels		−0.55	−1.55	0.46
				Digit span 1st error	Percentage correct		−0.32	−1.3	0.67
Tikuisis and Keefe (65)	1.12	180	**Friendly-foe target detection & marksmanship**	**Foe identification (Go)**	**Percentage correct**	**Positive**	**0.95**	**0.06**	**1.84**
				Friendly identification (no-go)	**Percentage correct**	**Negative**	**−1.33**	**−2.27**	**−0.39**
Caldwell et al. (57)	1.43	75	Shape recognition	Accuracy	Percentage correct		0.60	−0.41	1.6
			Shape + shade recognition	**Accuracy**	**Percentage correct**	**Positive**	**2.37**	**1.01**	**3.73**
			Reaction time	**Accuracy**	**Percentage correct**	**Positive**	**2.49**	**1.09**	**3.88**
Tamm et al. (63)	1.50	90	Time perception	Slope	—		−0.29	−0.92	0.33
				Intercept	—		0	−0.62	0.62
Radakovic et al. (2)	2.61	90	**Motor screening**	**Number of errors**	***Number incorrect**	**Negative**	**1.5**	**0.49**	**2.52**
			Reaction time	**Correct responses**	**Percentage correct**	**Negative**	**−2.25**	**−3.42**	**−1.08**
			Rapid visual information processing	**Correct responses**	**Percentage correct**	**Negative**	**−1.00**	**−1.94**	**−0.06**
Maric et al. (50)	2.72	90	Motor screening	Accuracy	Number correct		0.10	−0.52	0.72
			Reaction time	**Accuracy**	**Percentage correct**	**Negative**	**−0.89**	**−1.54**	**−0.23**
				Working Memory					
Saldaris et al. (15)	0.20	30	Serial sevens	Response rate	Count		0.17	−0.63	0.98
				Accuracy	Percentage correct		0.08	−0.72	0.88
Mazalan et al. (62)	0.46	60	Complex span task (OSPAN)	Accuracy	Number correct		−0.62	−1.53	0.28
			Serial sevens task	Accuracy	Number correct		−0.43	−1.32	0.46
Macleod et al. (6)	0.52	50	Sternberg 1 item	Accuracy	Percentage correct		0.17	−0.81	1.15
			Sternberg 3 item	Accuracy	Percentage correct		0	−0.98	0.98
			Sternberg 5 item	Accuracy	Percentage correct		−0.21	−1.19	0.78
			Visual search baseline	Accuracy	Percentage correct		0	0	0

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Table 3.— *Continued*

Reference	Peak T_{core} Difference, °C	Time Point	Cognitive Task	Performance Measure	Unit of Measure	Influence of Heat	Hedges' *g*	Lower CI	Upper CI
			Visual search complex	Accuracy	Percentage correct		0	−0.98	0.98
Clarke et al. (58)	0.53	15	Mathematical pairs	Accuracy	Percentage correct		−0.65	−1.48	0.17
Ashworth et al. (54)	1.08	15	**Memory**	**Accuracy**	**Number correct**	**Negative**	**−1.55**	**−2.71**	**−0.39**
			Digit span	Maximum span	Levels		−0.55	−1.55	0.46
				Digit span 1st error	Percentage correct		−0.32	−1.3	0.67
Deming et al. (49)	1.13	120	**Word list recall**	**Accuracy**	**Number correct**	**Negative**	**−1.27**	**−2.21**	**−0.34**
Maric et al. (50)	2.72	90	Matching to sample visual search	Accuracy	Percentage correct		−0.55	−1.18	0.09
			Pattern recognition memory (Spatial)	Accuracy	Percentage correct		−0.4	−1.03	0.23
			Pattern recognition memory (Visual)	Accuracy	Percentage correct		−0.35	−0.97	0.28
				Cognitive Flexibility					
Gerhart et al. (45)	0.10	30	Classical Stroop	Word correct responses	Number correct		—	—	—
				Color correct responses	Number correct		—	—	—
Saldaris et al. (15)	0.20	30	Color multi-source interference	Accuracy	Percentage correct		−0.07	−0.87	0.73
Edwards et al. (60)	0.43	95	Modified trail making	Digits per minute	Number correct		−0.27	−1.12	0.57
Wittbrodt et al. (64)	0.46	50	Trail making	Accuracy	Percentage correct		−0.18	−0.99	0.62
Donnan et al. (59)	0.48	40	Classical Stroop (Incongruent)	Accuracy	Percentage correct		−0.30	−1.10	0.51
			Classical Stroop (Congruent)	Accuracy	Percentage correct		−0.43	−1.24	0.38
Parker et al. (51)	0.50	90	**Wisconsin card sorting test**	**Perseverative response**	**Number correct**	**Negative**	**−0.88**	**−1.53**	**−0.23**
				Conceptual response	Number correct		0.44	−0.19	1.07
				Perseverative error	***Number incorrect**	**Positive**	**−1.01**	**−1.67**	**−0.35**
				Nonperseverative error	*Number incorrect		0.13	−0.49	0.75
Macleod et al. (6)	0.52	50	Classical Stroop (Incongruent)	Accuracy	Percentage correct		0.31	−0.67	1.3
			Classical Stroop (Congruent)	Accuracy	Percentage correct		0	−0.98	0.98
Ashworth et al. (54)	1.08	15	Trail making	Trail errors	*Number incorrect		0.46	−0.53	1.46
			Numerical Stroop	**Accuracy**	**Percentage correct**	**Negative**	**−2.07**	**−3.36**	**−0.79**
			Three-term reasoning	Accuracy	Percentage correct	Positive	1.8	0.58	3.01

Notes: For significant results, the influence of heat column indicates direction (negative is detrimental). "—", no available response; T_{core}, core temperature. Conditions at which peak changes in T_{core} occurred (between control and experimental) match the bolded conditions in Table 2. Hedges' *g* effect sizes: For incorrect (*) responses: Negative (−) indicates a performance benefit in experimental condition. For all other responses (i.e., percentage or number correct), a negative (−) indicates performance decline in experimental condition (less correct responses).

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Table 4. *Cognitive task results and core temperature changes for response times (displayed in ascending core temperature difference, for each cognitive domain)*

Reference	Peak T_{core} Difference, °C	Time Point	Cognitive Task	Performance Measure	Unit of Measure	Influence of Heat	Hedges' *g*	Lower CI	Upper CI
				Cognitive Inhibition					
Saldaris et al. (15)	0.20	30	Choice reaction time	Latency	Milliseconds		0.16	−0.64	0.96
Malan et al. (61)	0.25	70	Response time test	Response time	Milliseconds		0.70	−0.77	2.16
Saldaris et al. (21)	0.33	15	Choice reaction time	Reaction time	Milliseconds		−0.10	−0.98	0.78
				Movement time	Milliseconds		0	−0.88	0.88
Caldwell et al. (56)	0.43	150	Shape recognition	Response time	Milliseconds		0.79	−0.18	1.76
Donnan et al. (59)	0.48	40	Visual vigilance	Reaction time	Milliseconds		−0.09	−0.89	0.79
Aljaroudi et al. (53)	0.50	40	Go/no-go	Reaction time (go)	Milliseconds		0.08	−0.72	0.88
				Reaction time (no-go)	Milliseconds		0.20	−0.60	1.01
Parker et al. (51)	0.50	90	**Psychomotor vigilance test**	**Mean reaction time**	**Milliseconds**	**Positive**	**−4.92**	**−6.21**	**−3.61**
Shibasaki et al. (22)	0.67	60	Go/no-go	Reaction time	Milliseconds		0.02	−0.70	0.73
Clarke et al. (17)	0.71	30	Go/no-go	Response time	Milliseconds		−0.11	−1.09	0.87
Benjamin et al. (48)	0.92	75	Choice reaction time	Reaction time	Milliseconds		0	−0.80	0.80
Ashworth et al. (54)	1.08	15	Go/no-go	Cued reaction time	Milliseconds		−0.22	−1.20	0.76
				Un-cued reaction time	Milliseconds		−0.22	−1.20	0.77
			Perceptual processing	Global reaction time	Milliseconds		−0.33	−1.32	0.66
				Local reaction time	Milliseconds		−0.01	−0.99	0.97
				None, reaction time	Milliseconds		−0.42	−1.41	0.58
			Vigilance	**False starts**	**Milliseconds**	**Negative**	**1.12**	**0.04**	**2.19**
				Correct reaction time	Milliseconds		0.55	−0.45	1.55
Tikuisis and Keefe (65)	1.12	180	Friendly-foe target detection & marksmanship	Foe detection (go)	Milliseconds		−0.52	−1.37	0.33
				Friendly detection (no-go)	Milliseconds		−0.36	−1.2	0.48
Adams et al. (52)	1.46	90	Visual vigilance	Reaction time	Milliseconds		0	−0.80	0.80
Radakovic et al. (2)	2.61	90	Motor screening	Latency	Milliseconds		0.82	−0.1	1.74
			Reaction time	Reaction time	Milliseconds		0.18	−0.7	1.06
				Movement time	Milliseconds		0.23	−0.65	1.11
			Rapid visual information processing	**Latency**	**Milliseconds**	**Negative**	**0.63**	**0.27**	**1.53**
Maric et al. (50)	2.72	90	Motor screening	Latency	Milliseconds		0	−0.62	0.62
			Reaction time	Reaction time	Milliseconds		0.17	−0.45	0.79
				Movement time	Milliseconds		−0.17	−0.79	0.45
				Working Memory					
Wittbrodt et al. (64)	0.46	50	Pattern comparison	Reaction time	Milliseconds		0.33	−0.47	1.14
			Letter-digit recognition	Reaction time	Milliseconds		0.21	−0.59	1.01
Macleod et al. (6)	0.52	50	Sternberg 1 item	Response time	Milliseconds		−0.19	−1.17	0.79
			Sternberg 3 item	Response time	Milliseconds		−0.65	−1.66	0.37
			Sternberg 5 item	Response time	Milliseconds		−0.09	−1.07	0.89
Maric et al. (50)	2.72	90	Matching to sample visual search	Latency	Milliseconds		0.22	−0.41	0.84
			Pattern recognition memory (spatial)	Latency	Milliseconds		0.41	−0.22	1.03
			Pattern recognition memory (visual)	Latency	Milliseconds		0.16	−0.46	0.78

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Table 4.— *Continued*

Reference	Peak T_{core} Difference, °C	Time Point	Cognitive Task	Performance Measure	Unit of Measure	Influence of Heat	Hedges' *g*	Lower CI	Upper CI
Cognitive Flexibility									
Saldaris et al. (15)	0.20	30	Color multi-source interference	Latency	Milliseconds		0.05	−0.75	0.85
Donnan et al. (59)	0.48	40	**Classical Stroop (Incongruent)**	**Reaction time**	**Milliseconds**	**Positive**	**−0.35**	**−1.15**	**−0.23**
			Classical Stroop (Congruent)	**Reaction time**	**Milliseconds**	**Positive**	**−0.97**	**−1.82**	**−0.11**
Macleod et al. (6)	0.52	50	Classical Stroop (Incongruent)	Response time	Milliseconds		−0.01	−0.99	0.97
			Classical Stroop (Congruent)	Response time	Milliseconds		0.07	−0.91	1.05
Benjamin et al. (48)	0.92	75	Trail making A	Time to completion	Seconds		−0.25	−1.05	0.56
			Trail making B	Time to completion	Seconds		0.12	−0.69	0.92
Ashworth et al. (54)	1.08	15	Trail making A	Response time	Minutes		0	−0.98	0.98
			Trail making B	Response time	Minutes		0	−0.98	0.98
Deming et al. (49)	1.13	120	Classical Stroop (Incongruent)	Reaction time	Milliseconds		0.78	−0.1	1.66
			Classical Stroop (Congruent)	**Reaction time**	**Milliseconds**	**Negative**	**1.08**	**0.17**	**1.99**

For significant results, the influence of heat column indicates direction (negative is detrimental). "—", no available response; T_{core}, core temperature. Conditions at which peak changes in T_{core} occurred (between control and experimental) match the bolded conditions in Table 2. Hedges' g effect sizes: Negative (−) indicates a performance benefit in experimental condition. For significant results, the direction of the effect of heat conditions is dictated by the "effect of heat" column.

The two tasks that identified decrements in working memory accuracy were a standard memory task (54), and a word list recall task (49). Both studies produced moderate changes in core temperature, and so further investigation of these working memory tasks is required to confirm the validity and reliability of these outcomes.

Cognitive Flexibility

Cognitive flexibility is arguably the most complex of the three domains, requiring an individual to continually alter their perspective—either spatially, interpersonally, or in response to continually altering rules (41). Tasks of cognitive flexibility typically include those of grammatical reasoning and rule switching. Within the current review, seven discrete tasks were used across ten studies. Performance in both classical, and modified versions of a trail-making task (48, 54, 60, 64) remained consistently unaffected across multiple designs. Contradictory results were identified within classical Stroop (6, 49, 59) and Wisconsin card sorting tests (51). Singular uses of three term reasoning (54), numerical Stroop (54), and color multi-source interference (15) tasks prohibit any definitive conclusion being drawn about these tasks.

In the only use of the Wisconsin card sorting task (51), the accuracy of correct perseverative responses was detrimentally affected, whereas a beneficial effect of heat was identified for fewer perseverative errors. Although this study used a large sample size (n = 40), as noted in the original study, differences in cognitive performance between the control and experimental groups at baseline question the validity of these outcomes.

There were no significant changes in the response time (48, 54), accuracy, or error count (54, 64) outcomes, across multiple uses of both classical, and modified versions of a trail making task (60). Despite ideal experimental conditions of core temperature > 39°C (48), long exercise duration (48, 60, 64), and moderate increases in core temperature (54), all studies consisted of relatively small sample sizes ($n \leq 12$) and so appropriate power may not have been reached, potentially limiting the observation of any meaningful performance changes.

Classical Stroop task results were inconsistent. In two studies, there were no changes in congruent or incongruent accuracy (6, 49), or response time performance (6), when both studies incorporated a manipulation of hydration status. However, a positive effect of heat on both congruent and incongruent response times was observed in a standard heat manipulation design (59). Whereas a negative effect of heat on congruent (but not incongruent) response times (49) was identified with a twofold greater change in core temperature, and three times as long in exercise duration. The small to moderate temperature changes across tasks of cognitive flexibility potentially prevents observation of any notable changes in performance. Given the evidence to date, further assessments of Stroop and Wisconsin card sorting tasks are warranted.

Limitations and Future Directions

This review is not without limitations. For instance, the relatively small sample sizes (~16 participants per study), alongside the modest number of studies (n = 27), the authors were able to retrieve data from, and the limited number of

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cognitive tasks and time points from which to extract discrete data points from, meant it would have been inappropriate to perform a meta-analysis or meta-regression alongside the review (66). Although a meta-analysis may have produced statistically significant changes in cognitive performance, in the absence of a larger number of studies, the authors decided to avoid conducting a potentially misleading analysis through double-counting of studies, cognitive tasks, and raw data points, or reporting of a grouped effect size across multiple cognitive outcomes within any singular study (66–68). In addition, where cognitive performance is assessed during exercise interventions, often participants are asked to cease the activity, and in some instances, are moved to a thermoneutral room to perform the cognitive task. Subsequently, this can reduce both skin and core temperatures, which may affect the ability to accurately determine core temperature thresholds, when dropping during the assessment. Therefore, future studies may seek to abstain from moving participants to cooler environments during cognitive testing, as well as using short-duration cognitive batteries, to help maintain elevated core temperatures if the exercise has to be ceased to perform the tasks. Finally, while this review focused on the objective changes in core temperature in relation to cognitive performance changes, future research could seek to examine how such differences in cognitive assessments may correspond to subjective perceptions of exertion, and thermal strain.

CONCLUSIONS

This current review sought to examine the influence of exercise-induced heat strain on cognitive performance. Preliminary evidence did indicate that *1*) at least moderate levels of core temperature change (> 1°C) were required to influence cognitive performance across all three domains of cognitive performance (cognitive inhibition, working memory, and cognitive flexibility), and *2*) of the outcome measures evaluated, irrespective of cognitive domain, task accuracy was more detrimentally affected compared with task response time. However, of the utilized tasks, reaction time (cognitive inhibition), memory recall (working memory), and Stroop tasks (cognitive flexibility) appeared to be most likely to identify cognitive performance changes during heightened thermal strain.

A range of methodological discrepancies existed across research designs, including using multiple cognitive tasks of varying difficulties to manipulate cognitive load and experimental protocols, but there are also a range of extraneous factors that have the potential to confound the results. The present review has provided further evidence for the interplay of core temperature, physical activity, the external environment, and hydration status, on cognitive performance. Most notably, this review identified how changes in core temperature alone, are potentially not influential enough to produce consistent changes in cognitive performance, regardless of task complexity. Rather, a combination of factors is likely to heighten cognitive load beyond what core temperature alone would, and therefore, the cumulative influence of such factors—including exercise and dehydration—should be considered when analyzing the effect of heat strain on cognitive performance.

The present review highlighted how experimental designs that use short duration, intermittent protocols, commonly seen in team sports that allow recovery periods, rehydration, and where high core temperatures are not sustained for extended periods of time, may be unlikely to provoke significant decrements in cognitive performance. However, activities of a longer duration that do not allow as frequent rehydration strategies, and which drive core temperatures beyond 39°C, or require participants to function at an elevated core temperature for extended periods of time (>60 min), may be more likely to demonstrate a negative effect on cognitive performance, and, more specifically, impair the executive network function. The continual use of small sample sizes and inadequately powered experiments within the literature potentially prevents the statistical observance of small but meaningful changes in cognitive performance. Appropriate cognitive task selection and reduction of methodological discrepancies, encouragement of larger sample sizes, and promotion of consistent assessments and definitions of cognitive tasks and domains is essential for future studies to determine the true influence of exercise-induced heat strain on cognitive performance.

APPENDIX

PubMed

The following filters and search terms were used in PubMed for the literature search.

Filters: Humans, English

"Exercise"[Title/Abstract] OR "Exercise Test"[Title/Abstract] OR "Physical Exertion"[Title/Abstract] OR "Sports"[Title/Abstract] OR "Athletes"[Title/Abstract] OR "Exercise"[Mesh] OR "Exercise Movement Techniques"[Mesh] OR "Exercise Therapy"[Mesh] OR "Exercise Test"[Mesh] OR "Physical Exertion"[Mesh] OR "Sports"[Mesh] OR "Athletes"[Mesh]

AND

"Cognition"[Title/Abstract] OR "Executive Function"[Title/Abstract] OR "Performance, Psychomotor"[Title/Abstract] OR "Psychological Tests"[Title/Abstract] OR "Attention"[Title/Abs-tract] OR "Nervous System"[Title/Abstract] OR "Cognition"[Mesh] OR "Executive Function"[Mesh] "Learning"[Mesh] OR "Decision Making"[Mesh] OR "Psychological Tests"[Mesh] OR "Attention"[Mesh] OR "Nervous System"[Mesh] OR "Cognition Disorders"[Mesh] OR "Task Performance and Analysis"[Mesh]

AND

"Hot Temperature"[Title/Abstract] OR "Heat Stress Disorders"[Title/Abstract] OR "Hyperthermia, Induced"[Title/Abstract] OR "Extreme Heat"[Title/Abstract] OR "Extreme Hot Weather"[Title/Abstract] OR "Heat Stress"[Title/Abstract] OR "Hot Temperature"[Mesh] OR "Heat Stress Disorders"[Mesh] OR "Hyperthermia, Induced"[Mesh] OR "Extreme Hot Weather"[Mesh] OR "Heat-Shock Response"

Cinahl

The following filters and search terms were used in CINAHL for the literature search.

Filters: English Language, Peer Reviewed, Research Article, Human

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(TI (Exercise OR "Physical Activity" OR Fitness OR Workout OR "Exercise Test" OR "Exercise Physiology" OR "Exercise Tolerance" OR Sport OR Athletes OR "Work Capacity")) OR (AB (Exercise OR "Physical Activity" OR Fitness OR Workout OR "Exercise Test" OR "Exercise Physiology" OR "Exercise Tolerance" OR Sport OR Athletes OR 'Work Capacity')) OR (MH "Sports+") OR (MH "Physical Fitness+") OR (MH "Exercise+") OR (MH "Physical Activity") OR (MH "Physical Performance")

AND

(TI (Cognition OR "Cognitive Function" OR "Brain Function" OR Memory OR Brain OR "Cognitive Performance" OR "Cognitive Abilit*" OR "Executive Function*" OR Attention OR "Psychological Test*")) OR (AB (Cognition OR "Cognitive Function" OR "Brain Function" OR Memory OR Brain OR "Cognitive Performance" OR "Cognitive Abilit*" OR "Executive Function*" OR Attention OR "Psychological Test*")) OR (MH "Cognition+") OR (MH "Psychomotor Performance+") OR (MH "Neurocognitive (Iowa NOC)+") OR (MH "Neuropsychological Tests+") OR (MH "Memory+") OR (MH "Learning+") OR (MH "Mental Fatigue+") OR (MH "Academic Performance+") OR (MH "Brain+") OR (MH "Neuropsychology") OR (MH "Neural Pathways") OR ("MH Problem Solving+")

AND

(TI ("Heat Stress" OR "Thermal Stress" OR Heat OR "Hot Environment" OR "Heat Exposure" OR "Hot-Humid" OR "Warm Environment" OR "Heating" OR "Hyperthermi*" OR "Exertional Heat Stress" OR "Hot Conditions")) OR (AB ("Heat Stress" OR "Thermal Stress" OR Heat OR "Hot Environment" OR "Heat Exposure" OR "Hot-Humid" OR "Warm Environment" OR "Heating" OR "Hyperthermi*" OR "Exertional Heat Stress" OR "Hot Conditions")) OR (MH "Heat") OR (MH "Heat Stress Disorders+") OR (MH "Heating") OR (MH "Warming Techniques") OR (MH "Thermoregulation (Iowa NOC)") OR (MH "Body Temperature Regulation+") OR (MH "Hyperthermia, Induced+") OR (MH "Hyperthermia (Saba CCC)") OR (MH "Hyperthermia (NANDA)") OR (MH "Thermoregulation Impairment (Saba CCC)")

Embase

The following filters and search terms were used in Embase for the literature search.

Filters: Article, English, Human

("cognition"/exp OR "psychologic test"/exp OR "memory"/exp OR "job performance"/exp OR "academic achievement"/exp) OR cognition:ab,ti OR "cognitive function test":ab,ti OR "cognitive function":ab,ti OR "executive function":ab,ti OR memory:ab,ti OR "decision making":ab,ti OR attention:ab,ti OR "neuropsychological test":ab,ti OR "mental performance":ab,ti OR brain:ab,ti OR "task performance":ab,ti OR neuropsychology:ab,ti OR "academic achievement":ab,ti OR "psychomotor activity":ab,ti OR "psychomotor performance":ab,ti OR "psychologic test":ab,ti

AND

(("heat"/exp OR "temperature stress"/exp OR "thermal injury"/exp OR "high temperature"/exp OR "thermoregulation"/exp OR "thermogenesis"/exp) AND "thermoregulation"/exp OR "hyperthermia"/exp) OR "heat stress":ab,ti OR hyperthermia:ab,ti OR "high temperature":ab,ti OR "temperature stress":ab,ti OR heat:ab,ti OR thermoregulation:ab,ti OR "heat tolerance":ab,ti OR "heat exhaustion":ab,ti OR "heat stroke":ab,ti

AND

(("exercise"/exp OR "physical activity, capacity") AND "performance"/exp OR "fitness"/exp OR "exercise test"/exp OR "sport"/exp OR "athlete"/exp OR "work capacity"/exp OR "exercise physiology"/exp) OR (exercise:ab,ti OR "physical activity, capacity":ab,ti) AND performance:ab,ti OR "exercise test":ab,ti OR "exercise tolerance":ab,ti OR "exercise intensity":ab,ti OR "endurance training":ab,ti OR athlete:ab,ti OR fitness:ab,ti OR "exercise physiology":ab,ti OR "work capacity":ab,ti OR sport:ab,ti AND "article"/it AND [english]/lim AND "human"/de

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

M.A.D. conceived and designed research; M.A.D., J.T.W., and I.B.S. analyzed data; M.A.D. prepared figures; M.A.D., J.T.W., and I.B.S. drafted manuscript; M.A.D., J.T.W., N.K., M.J.W., and I.B.S. edited and revised manuscript; J.T.W. and I.B.S. approved final version of manuscript.

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